

Mathematical Foundations

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These notes develop the core mathematical toolkit for the course: the language of probability, three generating functions (MGF, CGF, characteristic function), Gaussian integration in arbitrary dimension, Wick's theorem, the saddle-point approximation, and the Hubbard–Stratonovich transformation. Every definition is followed by a worked example; every theorem is derived step by step.

Probability: basic definitions

A **sample space** Ω is the set of all possible outcomes of an experiment. A **random variable** X is a function $X : \Omega \rightarrow \mathbb{R}$ assigning a real number to each outcome. When X is continuous its distribution is described by a **probability density function** (pdf) $f_X(x)$ satisfying

$$f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1. \quad (1)$$

The probability that X falls in $[a, b]$ is $\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx$. The **cumulative distribution function** (CDF) is $F_X(x) = \int_{-\infty}^x f_X(t) dt$, so $f_X = F'_X$. When X is discrete we write $P(X = x_k) = p_k$ with $\sum_k p_k = 1$.

The **expectation** and **variance** are

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx, \quad \text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2. \quad (2)$$

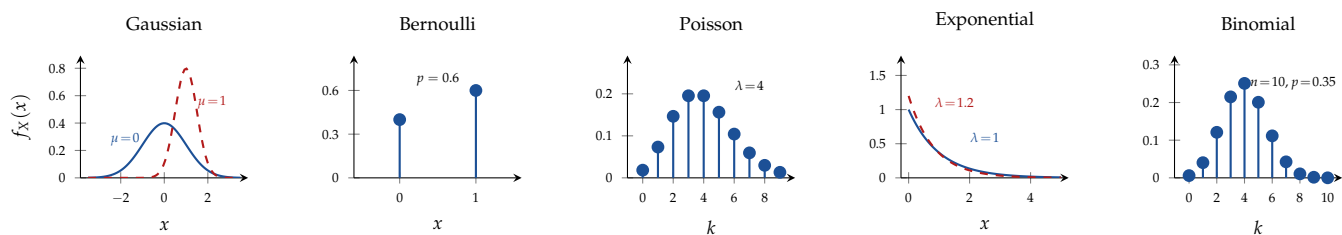


Figure 1: The five distributions introduced in this section. *Blue*: primary parameter choice; *dashed red*: alternative parameter.

Classic distributions

THE FIVE DISTRIBUTIONS shown in Figure 1 appear throughout the course.

Gaussian. $X \sim \mathcal{N}(\mu, \sigma^2)$ has pdf

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}. \quad (3)$$

It maximises entropy subject to fixed mean and variance, and is the attractor for sums of finite-variance variables (CLT).

Bernoulli. $X \sim \text{Bernoulli}(p)$: $P(X = x) = p^x(1-p)^{1-x}$, $x \in \{0, 1\}$. Mean p , variance $p(1-p)$.

Binomial. $X \sim \text{Binomial}(n, p)$: $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$. Mean np , variance $np(1-p)$.

Poisson. $X \sim \text{Poisson}(\lambda)$: $P(X = k) = e^{-\lambda} \lambda^k / k!$. Mean = variance = λ .

Exponential. $X \sim \text{Exp}(\lambda)$: $f_X(x) = \lambda e^{-\lambda x}$, $x \geq 0$. The only continuous memoryless distribution.

Generating functions

THREE GENERATING FUNCTIONS encode all statistical information in a distribution. They are the main calculational tools we will use to manipulate and identify distributions.

Moment generating function

The **moment generating function** (MGF) is

$$M_X(t) = \mathbb{E}[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx. \quad (4)$$

Expanding $e^{tX} = \sum_{n=0}^{\infty} (tX)^n / n!$ and taking expectations term by term,

$$M_X(t) = \sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbb{E}[X^n], \quad (5)$$

so the n -th moment is recovered by differentiation:

$$\mathbb{E}[X^n] = \left. \frac{d^n M_X}{dt^n} \right|_{t=0}. \quad (6)$$

For independent X, Y , the MGF of the sum factorises: $M_{X+Y}(t) = M_X(t)M_Y(t)$, since $\mathbb{E}[e^{t(X+Y)}] = \mathbb{E}[e^{tX}]\mathbb{E}[e^{tY}]$ under independence.

MGF of the Gaussian

For $X \sim \mathcal{N}(\mu, \sigma^2)$:

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx. \quad (7)$$

Combine the exponentials:

$$tx - \frac{(x - \mu)^2}{2\sigma^2} = -\frac{(x - (\mu + \sigma^2 t))^2}{2\sigma^2} + \mu t + \frac{\sigma^2 t^2}{2}. \quad (8)$$

The constant terms factor out; the remaining integral is a Gaussian normalisation equal to 1:

$$M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right). \quad (9)$$

Differentiating at $t = 0$ gives $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \sigma^2$.

MGF of the Bernoulli

$M_X(t) = (1 - p) + pe^t$. Then $M'_X(0) = p$ and $\text{Var}(X) = M''_X(0) - (M'_X(0))^2 = p(1 - p)$.

Cumulant generating function

The **cumulant generating function** (CGF) is

$$K_X(t) = \log M_X(t), \quad (10)$$

and the n -th **cumulant** is $\kappa_n = K_X^{(n)}(0)$. The first two are $\kappa_1 = \mathbb{E}[X]$ and $\kappa_2 = \text{Var}(X)$; higher cumulants measure non-Gaussianity (κ_3 is skewness, κ_4 is excess kurtosis). All cumulants of a Gaussian beyond the second vanish.

WHY THE LOGARITHM? Because for independent X, Y ,

$$K_{X+Y}(t) = K_X(t) + K_Y(t). \quad (11)$$

Cumulants are additive for independent variables; moments are not. This makes the CGF the natural tool for large-system analysis.

CGF of the Gaussian

$K_X(t) = \mu t + \frac{1}{2}\sigma^2 t^2$. The CGF is a polynomial of degree 2 — the hallmark of the Gaussian. Hence $\kappa_1 = \mu$, $\kappa_2 = \sigma^2$, $\kappa_n = 0$ for $n \geq 3$.

CGF of the Poisson

The MGF is $M_X(t) = e^{-\lambda} \sum_k (\lambda e^t)^k / k! = \exp(\lambda(e^t - 1))$, so

$$K_X(t) = \lambda(e^t - 1), \quad \kappa_n = \lambda \quad \forall n \geq 1. \quad (12)$$

All cumulants equal λ , confirming $\mathbb{E}[X] = \text{Var}(X) = \lambda$.

Characteristic function

The **characteristic function** (Fourier transform of f_X) is

$$\phi_X(t) = \mathbb{E}[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} f_X(x) dx, \quad t \in \mathbb{R}. \quad (13)$$

It always exists since $|e^{itx}| = 1$, and it *uniquely* determines the distribution even when the MGF does not exist (heavy tails). The inversion formula reconstructs the pdf:

$$f_X(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \phi_X(t) dt. \quad (14)$$

Gaussian. $\phi_X(t) = \exp(i\mu t - \frac{1}{2}\sigma^2 t^2)$, obtained by substituting $t \rightarrow it$ in (9).

Figure 2 summarises the relationships among the three generating functions.

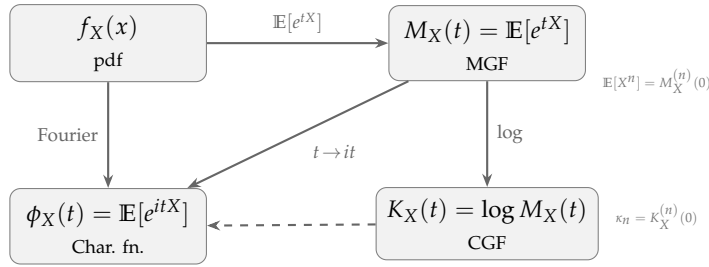


Figure 2: Relationships among the three generating functions. Arrows show how each is constructed from the previous.

Multivariate distributions

FOR TWO RANDOM VARIABLES X, Y the **joint pdf** $f_{X,Y}(x, y)$ satisfies $\iint f_{X,Y} dx dy = 1$. The **marginal** pdfs are

$$f_X(x) = \int f_{X,Y}(x, y) dy, \quad f_Y(y) = \int f_{X,Y}(x, y) dx. \quad (15)$$

The **conditional** pdf is $f_{X|Y}(x|y) = f_{X,Y}(x, y) / f_Y(y)$, which is Bayes' theorem in density form: $f_{X|Y}(x|y) \propto f_{Y|X}(y|x) f_X(x)$.

X and Y are **independent** iff $f_{X,Y} = f_X f_Y$. The **covariance** and **correlation** are

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y], \quad \rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \in [-1, 1]. \quad (16)$$

The multivariate Gaussian

A random vector $\mathbf{x} \in \mathbb{R}^N$ follows $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C})$ if

$$f(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^N \det \mathbf{C}}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{C}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right). \quad (17)$$

We derive this form from first principles in Section .

Gaussian integrals

GAUSSIAN INTEGRALS are the fundamental computational engine of statistical physics and machine learning theory. The ability to evaluate them in closed form — even in hundreds of dimensions — is what makes Gaussian models analytically tractable.

The one-dimensional integral

$$\int_{-\infty}^{\infty} e^{-\frac{1}{2}ax^2} dx = \sqrt{\frac{2\pi}{a}}, \quad a > 0. \quad (18)$$

Derivation. Let $I = \int e^{-\frac{a}{2}x^2} dx$ and square it:

$$I^2 = \iint e^{-\frac{a}{2}(x^2+y^2)} dx dy. \quad (19)$$

Switch to polar coordinates $x = r \cos \theta$, $y = r \sin \theta$, with area element $r dr d\theta$:

$$I^2 = 2\pi \int_0^{\infty} r e^{-\frac{a}{2}r^2} dr. \quad (20)$$

Substitute $u = \frac{a}{2}r^2$ so $r dr = du/a$:

$$I^2 = \frac{2\pi}{a} \int_0^{\infty} e^{-u} du = \frac{2\pi}{a}, \quad (21)$$

giving $I = \sqrt{2\pi/a}$. The polar-coordinates trick is illustrated in the margin.

Adding a linear source term by completing the square, $-\frac{a}{2}x^2 + bx = -\frac{a}{2}(x - b/a)^2 + b^2/(2a)$, and shifting $x \rightarrow x + b/a$:

$$\int_{-\infty}^{\infty} e^{-\frac{1}{2}ax^2+bx} dx = \sqrt{\frac{2\pi}{a}} e^{\frac{b^2}{2a}}. \quad (22)$$

The multidimensional integral

Let $\mathbf{x} \in \mathbb{R}^N$, $\mathbf{A}$ real symmetric positive-definite, $\mathbf{b} \in \mathbb{R}^N$. Then:

$$\int_{\mathbb{R}^N} d^N x \exp\left(-\frac{1}{2}\mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x}\right) = \sqrt{\frac{(2\pi)^N}{\det \mathbf{A}}} \exp\left(\frac{1}{2}\mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}\right). \quad (23)$$

Step 1: Complete the square. Claim:

$$-\frac{1}{2}\mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} = -\frac{1}{2}(\mathbf{x} - \mathbf{A}^{-1}\mathbf{b})^T \mathbf{A}(\mathbf{x} - \mathbf{A}^{-1}\mathbf{b}) + \frac{1}{2}\mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}. \quad (24)$$

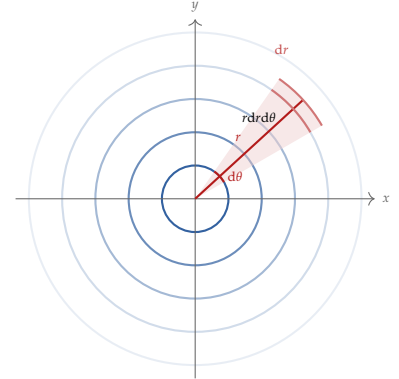


Figure 3: Squaring I converts the product of two 1-D Gaussians into a 2-D radially symmetric integral. The area element in polar coordinates is $r dr d\theta$.

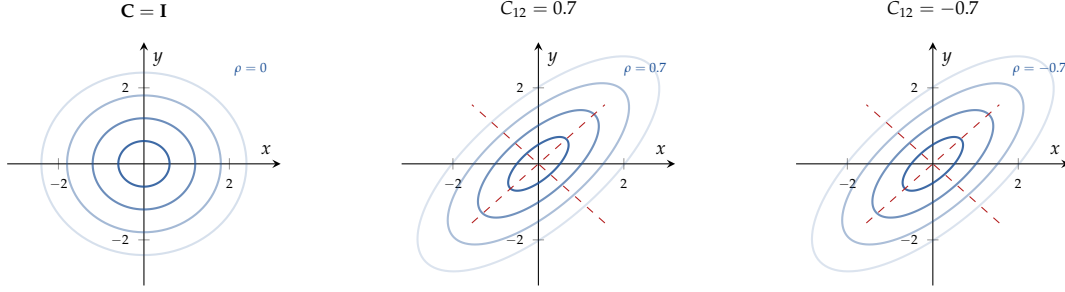


Figure 4: Level sets (constant-density contours) of 2-D Gaussians. Left: isotropic ($C = I$), circular contours. Centre: positive correlation $\rho = 0.7$, ellipse tilted at 45° . Right: negative correlation $\rho = -0.7$, ellipse tilted at -45° . Dashed red lines mark the principal axes (eigenvectors of C).

Expanding the right-hand side and using $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$ and symmetry of $\mathbf{A}$:

$$\begin{aligned} \text{RHS} &= -\frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} + \frac{1}{2}\mathbf{x}^T\mathbf{b} + \frac{1}{2}\mathbf{b}^T\mathbf{x} \\ &\quad - \frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b} + \frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b} \\ &= -\frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} + \mathbf{b}^T\mathbf{x}, \end{aligned} \quad (25)$$

confirming (24).

Step 2: Shift the variable. Set $\mathbf{y} = \mathbf{x} - \mathbf{A}^{-1}\mathbf{b}$ ($d^N y = d^N x$):

$$\int d^N x e^{-\frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} + \mathbf{b}^T\mathbf{x}} = e^{\frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b}} \int d^N y e^{-\frac{1}{2}\mathbf{y}^T\mathbf{A}\mathbf{y}}. \quad (26)$$

Step 3: Diagonalise $\mathbf{A}$. By the spectral theorem $\mathbf{A} = \mathbf{O}^T\mathbf{\Lambda}\mathbf{O}$ with $\mathbf{O}$ orthogonal and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_N)$, $\lambda_i > 0$. Set $\mathbf{u} = \mathbf{O}\mathbf{y}$, so $d^N u = d^N y$ and $\mathbf{y}^T\mathbf{A}\mathbf{y} = \sum_i \lambda_i u_i^2$. The integral factorises:

$$\int d^N y e^{-\frac{1}{2}\mathbf{y}^T\mathbf{A}\mathbf{y}} = \prod_{i=1}^N \sqrt{\frac{2\pi}{\lambda_i}} = \sqrt{\frac{(2\pi)^N}{\det \mathbf{A}}}. \quad (27)$$

Combining (26) and (27) gives (23).

The source trick and moment generation

Define the **generating functional**¹

$$Z(\mathbf{b}) = \int d^N x e^{-\frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} + \mathbf{b}^T\mathbf{x}} = Z_0 e^{\frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b}}, \quad (28)$$

where $Z_0 = Z(0) = \sqrt{(2\pi)^N / \det \mathbf{A}}$. Since $\partial e^{\mathbf{b}^T\mathbf{x}} / \partial b_i = x_i e^{\mathbf{b}^T\mathbf{x}}$, differentiating and setting $\mathbf{b} = \mathbf{0}$ extracts moments:

$$\langle x_i \rangle = \frac{1}{Z_0} \frac{\partial Z}{\partial b_i} \Big|_{\mathbf{b}=\mathbf{0}}. \quad (29)$$

First moment. $\partial Z / \partial b_i = Z(\mathbf{b})(\mathbf{A}^{-1}\mathbf{b})_i$. At $\mathbf{b} = \mathbf{0}$: $\langle x_i \rangle = 0$.

Second moment. Differentiating again:

$$\frac{\partial^2 Z}{\partial b_i \partial b_j} \Big|_{\mathbf{b}=\mathbf{0}} = Z_0 (\mathbf{A}^{-1})_{ij}, \quad (30)$$

¹ The name *generating functional* comes from field theory; here it is simply the Gaussian integral as a function of the source $\mathbf{b}$.

so

$$\langle x_i x_j \rangle = (\mathbf{A}^{-1})_{ij} = C_{ij}. \quad (31)$$

The two-point correlator equals the (i, j) entry of the inverse of the matrix in the exponent. We call C_{ij} the **propagator**. The level sets of 2-D Gaussians for different covariance structures are shown in Figure 4.

Wick's theorem

WICK'S THEOREM gives an algorithmic rule for computing any moment of a Gaussian in terms of the propagator alone. It is the central combinatorial identity of perturbative field theory.

Statement

Let $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$. Then:

- (1) All odd moments vanish.
- (2) All even moments satisfy:

$$\langle x_{i_1} x_{i_2} \cdots x_{i_{2n}} \rangle = \sum_{\text{pairings}} \prod_{\text{pairs } (j,k)} C_{i_j i_k}, \quad (32)$$

where the sum runs over all $(2n - 1)!! = 1 \cdot 3 \cdot 5 \cdots (2n - 1)$ distinct ways to pair $\{i_1, \dots, i_{2n}\}$ into n unordered pairs.

Derivation via the generating functional

With $Z(\mathbf{b}) = Z_0 e^{\frac{1}{2} \mathbf{b}^T \mathbf{C} \mathbf{b}}$, expand:

$$e^{\frac{1}{2} \mathbf{b}^T \mathbf{C} \mathbf{b}} = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{1}{2} \sum_{i,j} b_i C_{ij} b_j \right)^n. \quad (33)$$

The m -th moment is $\langle x_{i_1} \cdots x_{i_m} \rangle = \partial^m e^{\frac{1}{2} \mathbf{b}^T \mathbf{C} \mathbf{b}} / \partial b_{i_1} \cdots \partial b_{i_m} |_{\mathbf{b}=\mathbf{0}}$.

For odd $m = 2n + 1$: every term in (33) has even powers of $\mathbf{b}$; after $2n + 1$ derivatives at least one b factor remains, which vanishes at $\mathbf{0}$. All odd moments vanish.

For even $m = 2n$: only the n -th term in (33) survives. Expanding it,

$$\frac{1}{n! 2^n} \sum_{i_1, j_1} \cdots \sum_{i_n, j_n} C_{i_1 j_1} \cdots C_{i_n j_n} b_{i_1} b_{j_1} \cdots b_{i_n} b_{j_n}. \quad (34)$$

Differentiating $2n$ times matches each b_{k_l} to exactly one index; the $\frac{1}{n! 2^n}$ prefactor precisely cancels the $n! 2^n$ over-counting of pairings, leaving the sum over distinct pairings of products of $C_{k_a k_b}$. This is Eq. (32).

Worked examples

Fourth moment

For $X \sim \mathcal{N}(0, \sigma^2)$ there are $(4 - 1)!! = 3$ pairings of $\{1, 2, 3, 4\}$:

- $(1, 2)(3, 4)$: contributes $\sigma^2 \cdot \sigma^2 = \sigma^4$.
- $(1, 3)(2, 4)$: contributes σ^4 .
- $(1, 4)(2, 3)$: contributes σ^4 .

So $\langle X^4 \rangle = 3\sigma^4$. (The three pairings are illustrated in the margin.)

Verification via the MGF: differentiating $e^{\frac{1}{2}\sigma^2 t^2}$ four times at $t = 0$ gives $3\sigma^4$. ✓

Two-variable moment

For $(x_1, x_2) \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$, the three pairings of $(1, 1, 2, 2)$ give

$$\langle x_1^2 x_2^2 \rangle = C_{11}C_{22} + 2C_{12}^2. \quad (35)$$

For independent variables ($C_{12} = 0$) this reduces to $\langle x_1^2 \rangle \langle x_2^2 \rangle$. ✓

Sixth moment

$(6 - 1)!! = 15$ pairings, each contributing σ^6 , so $\langle X^6 \rangle = 15\sigma^6$.

Saddle-point approximation

MANY INTEGRALS in statistical physics and machine learning take the form

$$I(N) = \int_{-\infty}^{\infty} dx e^{-Nf(x)}, \quad (36)$$

where N is a large parameter (number of neurons, spins, data points) and $f(x)$ is smooth. As $N \rightarrow \infty$, the integrand concentrates exponentially near the minimum of f .

One-dimensional case

Let x^* minimise f : $f'(x^*) = 0$, $f''(x^*) > 0$. Taylor expand:

$$f(x) = f(x^*) + \underbrace{f'(x^*)}_{0}(x - x^*) + \frac{f''(x^*)}{2}(x - x^*)^2 + \dots \quad (37)$$

Substituting and evaluating the Gaussian integral:

$$I(N) \approx \sqrt{\frac{2\pi}{Nf''(x^*)}} e^{-Nf(x^*)}, \quad N \rightarrow \infty. \quad (38)$$

The cubic correction is $\mathcal{O}(N^{-1/2}) \rightarrow 0$ since fluctuations are of order $(x - x^*) \sim N^{-1/2}$.

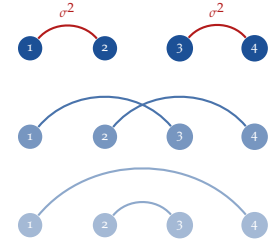


Figure 5: The three Wick pairings of $\{1, 2, 3, 4\}$ for computing $\langle X^4 \rangle$. Each arc contributes a propagator σ^2 .

Saddle-point concentration

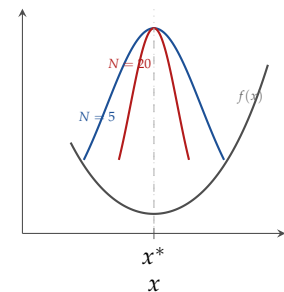


Figure 6: As N grows, $e^{-Nf(x)}$ concentrates ever more sharply near x^* . The saddle-point approximation replaces f with its quadratic expansion around x^* .

Multidimensional case

For $\mathbf{x}^*$ with $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and Hessian $\mathbf{H}_{ij} = \partial^2 f / \partial x_i \partial x_j|_{\mathbf{x}^*}$:

$$\int_{\mathbb{R}^N} d^N x e^{-Nf(\mathbf{x})} \approx \left(\frac{2\pi}{N} \right)^{N/2} \frac{e^{-Nf(\mathbf{x}^*)}}{\sqrt{\det \mathbf{H}(\mathbf{x}^*)}}. \quad (39)$$

Example: Stirling's approximation

From $n! = \int_0^\infty x^n e^{-x} dx$, substitute $x = nu$:

$$n! = n^{n+1} \int_0^\infty e^{-n(u - \log u)} du. \quad (40)$$

Identify $f(u) = u - \log u$. Saddle: $f'(1) = 0$, $f''(1) = 1$, $f(1) = 1$. Applying (38):

$$\log n! \approx n \log n - n + \frac{1}{2} \log(2\pi n). \quad (41)$$

Example: Entropy of a spin system

The number of configurations of N spins with magnetisation $m = \frac{1}{N} \sum_i \sigma_i$ is $\Omega(m) = \binom{N}{N(1+m)/2}$. Using Stirling with $n_\pm = N(1 \pm m)/2$:

$$s(m) \equiv \frac{\log \Omega}{N} = -\frac{1+m}{2} \log \frac{1+m}{2} - \frac{1-m}{2} \log \frac{1-m}{2}. \quad (42)$$

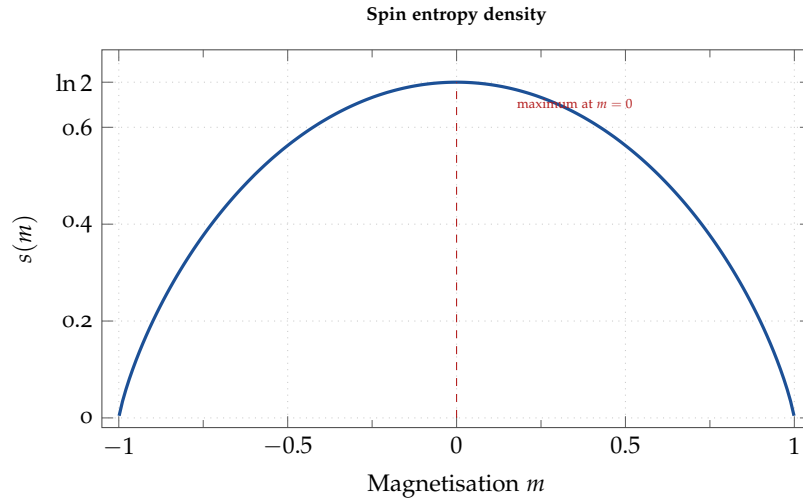


Figure 7: The spin entropy density $s(m)$ from Eq. (42). It is maximised at $m = 0$ (maximum disorder, $s = \log 2$) and vanishes at $m = \pm 1$ (all spins aligned).

Hubbard–Stratonovich transformation

THE HUBBARD–STRATONOVICH (HS) TRANSFORMATION is the central algebraic trick of mean-field theory. It converts an intractable quadratic coupling between many degrees of freedom into an integral over a single auxiliary field, making the saddle-point approximation applicable.

The basic identity

$$e^{\frac{a^2}{2}} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dz e^{-\frac{z^2}{2} + az}. \quad (43)$$

Proof. Complete the square in z : $-\frac{z^2}{2} + az = -\frac{1}{2}(z - a)^2 + \frac{a^2}{2}$. Then:

$$\frac{1}{\sqrt{2\pi}} \int e^{-\frac{z^2}{2} + az} dz = \frac{e^{a^2/2}}{\sqrt{2\pi}} \int e^{-\frac{(z-a)^2}{2}} dz = \frac{e^{a^2/2}}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = e^{a^2/2}. \quad \checkmark \quad (44)$$

General rescaled form

For $\alpha > 0$ and coupling J , substituting $a = J/\sqrt{\alpha}$ and $z \rightarrow \sqrt{\alpha}w$ in (43):

$$e^{\frac{J^2}{2\alpha}} = \sqrt{\frac{\alpha}{2\pi}} \int_{-\infty}^{\infty} dz e^{-\frac{\alpha}{2}z^2 + Jz}. \quad (45)$$

Derivation. Start from (43) with $a = J/\sqrt{\alpha}$. Substitute $z = \sqrt{\alpha}w$:

$$e^{\frac{J^2}{2\alpha}} = \frac{1}{\sqrt{2\pi}} \int e^{-\frac{z^2}{2} + \frac{Jz}{\sqrt{\alpha}}} dz = \frac{\sqrt{\alpha}}{\sqrt{2\pi}} \int e^{-\frac{\alpha w^2}{2} + Jw} dw. \quad (46)$$

Decoupling pairwise interactions

The key application: a Boltzmann weight $e^{\frac{\beta J}{2N}(\sum_i \sigma_i)^2}$ couples every pair of spins. Setting $\alpha = N\beta J$ and $J_{\text{eff}} = \beta J \sum_i \sigma_i$ in (45):

$$e^{\frac{\beta J}{2N}(\sum_i \sigma_i)^2} = \sqrt{\frac{N\beta J}{2\pi}} \int dm \exp\left(-\frac{N\beta J}{2}m^2 + \beta Jm \sum_{i=1}^N \sigma_i\right). \quad (47)$$

After this transformation the exponent is *linear* in each σ_i : the spins see only the uniform external field βJm and are completely decoupled from one another.

Application: Curie–Weiss mean-field theory

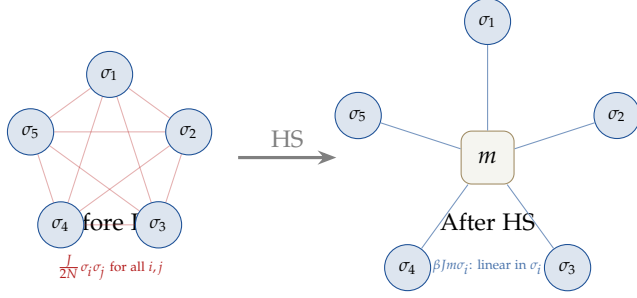


Figure 8: The Hubbard–Stratonovich transformation. Left: the original problem has all-to-all quadratic interactions $J\sigma_i\sigma_j/2N$, making the partition function a sum of 2^N entangled terms. Right: after introducing the auxiliary field m , each spin interacts only with m through a *linear* coupling $\beta J m \sigma_i$; the spins decouple and the sum factorises.

THE CURIE–WEISS MODEL is the fully-connected Ising model:

$$H = -\frac{J}{2N} \sum_{i,j} \sigma_i \sigma_j = -\frac{J}{2N} \left(\sum_i \sigma_i \right)^2, \quad \sigma_i \in \{+1, -1\}. \quad (48)$$

The partition function is $Z = \sum_{\{\sigma_i\}} e^{-\beta H}$.

Step 1 — Apply HS.

$$Z = \sqrt{\frac{N\beta J}{2\pi}} \int dm e^{-\frac{N\beta J}{2} m^2} \sum_{\{\sigma_i\}} \prod_{i=1}^N e^{\beta J m \sigma_i}. \quad (49)$$

Step 2 — Factorise. Spins are independent after HS:

$$\sum_{\{\sigma_i\}} \prod_i e^{\beta J m \sigma_i} = \prod_i (e^{\beta J m} + e^{-\beta J m}) = (2 \cosh \beta J m)^N. \quad (50)$$

Step 3 — Write as single integral.

$$Z = \sqrt{\frac{N\beta J}{2\pi}} \int dm \exp(-Nf(m)), \quad f(m) = \frac{\beta J}{2} m^2 - \log(2 \cosh \beta J m). \quad (51)$$

Step 4 — Saddle-point. $f'(m^*) = \beta J [m^* - \tanh(\beta J m^*)] = 0$:

$$m^* = \tanh(\beta J m^*). \quad (52)$$

This is the **mean-field self-consistency equation**. The auxiliary field m is the magnetisation $m^* = \langle \sigma_i \rangle$.

Step 5 — Phase structure. Non-trivial solutions $m^* \neq 0$ require the slope of $\tanh(\beta J m)$ at $m = 0$ to exceed 1:

$$\beta J > 1 \iff T < T_c \equiv J. \quad (53)$$

Above T_c only the paramagnetic solution $m^* = 0$ exists. Below T_c two ferromagnetic solutions $m^* = \pm m_0$ appear — this is the **ferromagnetic phase transition**, illustrated in Figure 9. The entire derivation (HS $\rightarrow$ spin factorisation $\rightarrow$ saddle-point $\rightarrow$ self-consistency equation) is the prototype for every mean-field calculation in this course, including the Hopfield model and the replica method.

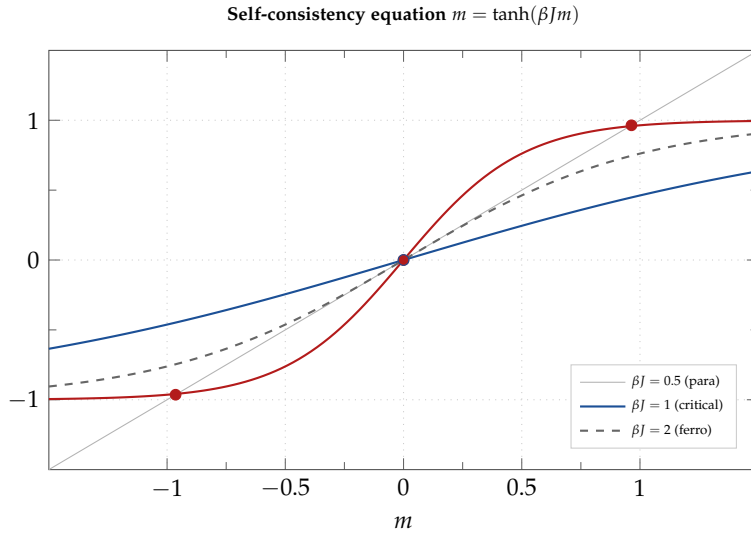


Figure 9: Graphical solution of the self-consistency equation (52). Intersections of $y = \tanh(\beta J m)$ with the diagonal $y = m$ give the solutions m^* . Below T_c ($\beta J > 1$, red curve) three intersections appear: $m^* = 0$ (unstable) and $m^* = \pm m_0$ (stable, filled dots). Above T_c ($\beta J < 1$, blue) only $m^* = 0$ exists.